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Quercetin analogs as α-glucosidase inhibitors with antidiabetic activity

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ABSTRACT

This study investigated the inhibitory effects of synthesized quercetin analogs (Q1–Q23) on α-glucosidase, particularly Q7. Q7, a synthetic analog of quercetin, exhibits superior α-glucosidase inhibitory properties compared with quercetin and the standard drug acarbose. This effect is attributed to the unique $-CF_3$ group on the benzene ring of Q7, which enhances its inhibitory effect. As a non-competitive inhibitor, Q7 shows strong binding and stability within the α-glucosidase-Q7 complex. *In vivo* experiments showed that Q7 significantly reduced fasting blood sugar levels in diabetic mice, surpassing the effects of metformin. In addition, it affects the gut microbiota, particularly the microbes associated with anti-inflammatory responses and short-chain fatty acid production, which are crucial for the treatment of type 2 diabetes. However, the current Q7 dosage led to carbohydrate accumulation in the colon of mice, impairing the function of the intestinal microbiota. This study highlights the potential of Q7 for treating type 2 diabetes and emphasizes the need to determine the optimal dosages that balance therapeutic effects and gut health.

1. Introduction

Diabetes mellitus, a global health threat, is the ninth leading cause of death, affecting one in ten adults worldwide (Yoshida et al., 2008). Chronic metabolic diseases are characterized by high blood sugar levels owing to inadequate insulin production. Type 1 diabetes mellitus (T1DM) is characterized by the immune-mediated destruction of insulin-producing cells, whereas type 2 diabetes mellitus (T2DM) is associated with insulin resistance and is influenced by genetic, lifestyle, and environmental factors (Ding et al., 2022; Li et al., 2019). T2DM can lead to complications such as cardiovascular disease, neuropathy, and kidney damage. Inhibition of α-glucosidase is an effective treatment for T2DM (Luo et al., 2023; Santos et al., 2018).

α-Glucosidase (EC 3.2.1.20) is an exotype and the key catabolic enzyme of carbohydrates, known as hydrolase. It is a membrane-bound enzyme in the small intestine that catalyzes the conversion of non-absorbable di- and oligosaccharides into absorbable monosaccharide glucose molecules. The inhibition of α-glucosidase is a key strategy to treat disorders related to carbohydrate uptake, including diabetes, obesity, dental caries, and periodontal diseases (Akhter et al., 2021; Ullah et al., 2023). However, approved α-glucosidase inhibitors such as acarbose, miglitol, and voglibose have undesirable gastrointestinal side effects that hinder their use (Mushtaq et al., 2023; Popović et al., 2023; Santos et al., 2022; Taha et al., 2021). Therefore, new drugs to treat diabetes and prevent or treat complications are urgently needed.

Quercetin and cinnamic acid stand out among natural products because of their ability to inhibit α-glucosidase. Quercetin is characterized by various mechanisms such as inhibition of glucose absorption, improvement of insulin sensitivity, and antioxidant and anti-inflammatory effects (Zhang et al., 2021). This versatility helps to stabilize blood sugar levels and relieves postprandial hyperglycemia. In contrast, cinnamic acid exhibits potent hypoglycemic properties by improving the insulin response, reducing insulin resistance, slowing carbohydrate absorption, and exhibiting anti-inflammatory and antioxidant properties (Yu et al., 2023). Both these compounds are critical for protecting insulin production and minimizing the negative effects of

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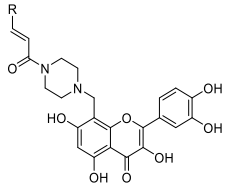
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Table 1
Inhibitory activities of the target compounds for α-glucosidase.



Species	R	IC_{50}(μM)[a]		R	IC_{50}(μM)[a]
Q1		2.07 ± 0.29	**Q13**		7.86 ± 1.37
Q2		1.70 ± 0.23	**Q14**		2.63 ± 0.04
Q3		3.17 ± 0.13	**Q15**		2.96 ± 0.07
Q4		3.38 ± 0.02	**Q16**		2.08 ± 0.65
Q5		3.28 ± 0.27	**Q17**		3.82 ± 0.36
Q6		2.05 ± 0.71	**Q18**		2.83 ± 0.03
Q7		1.06 ± 0.38	**Q19**		5.30 ± 0.35
Q8		2.68 ± 0.19	**Q20**		3.65 ± 0.87
Q9		2.79 ± 0.36	**Q21**		2.84 ± 0.21
Q10		2.03 ± 0.21	**Q22**		1.67 ± 0.53
Q11		2.60 ± 0.64	**Q23**		5.97 ± 0.01
Q12		2.95 ± 0.64	**Quercetin**		7.97 ± 0.89
Acarbose		124.22 ± 4.48			

[a] Values are mean ± SD. All experiments were performed at least three times.

chronic inflammation and oxidative stress in regulating blood sugar. By combining quercetin and cinnamic acid, we synthesized a new molecule that is expected to have an enhanced inhibitory effect. In addition, we simulated molecular docking and molecular dynamics by measuring the enzyme inhibition kinetics using Lineweaver–Burk plots to predict the target of action of these agents on α-glucosidase, and validated our results through *in vivo* experiments in mice. This comprehensive series of studies is not only important in the field of nutrition, but also holds great potential in the search for innovative therapeutic approaches and medications in clinical practice.

2. Materials and methods

2.1. Materials

α-Glucosidase, extracted from *Saccharomyces cerevisiae* (EC3.2.1.20), was obtained from Sigma-Aldrich (St. Louis, MO, USA). The chemicals and solvents were commercially available and used without further purification. ^{1}H NMR and ^{13}C NMR spectra were recorded on a Bruker 400 MHz at 25 °C with TMS as an internal standard in the deuterium reagent. The MS analysis was performed using an LTQ Orbitrap Elite instrument (Thermo Fisher Scientific, Waltham, MA, USA).

2.2. α-Glucosidase inhibitory activity

The α-glucosidase inhibitory activity of quercetin analogs was evaluated as described in a previous report with modifications (Taj et al., 2022; Zhang et al., 2014). In the reaction, 30 μL of α-glucosidase (0.5 U/mL in 0.1 mol/L PBS) and 120 μL of 0.1 mol/L PBS (pH 6.8) were mixed with 10 μL of the test sample and incubated for 10 min at 37 °C. Then, 80 μL PNPG (2.5 mM) was added, and the mixture was incubated for another 10 min at 37 °C. The absorbance was measured at 405 nm using an enzyme marker. Acarbose served as the positive control, DMSO was used as the negative control, and a solution without substrate was used as a blank control.

2.3. Kinetics of enzyme inhibition

The enzyme kinetics of compounds Q7, Q13, Q22, and acarbose toward α-glucosidase were determined according to a previously described method. Inhibition kinetics were analyzed using different concentrations of compounds Q7, Q13, Q22, and acarbose (0–250 μM) and different substrate concentrations (0–2.4 mM). The Lineweaver–Burk plot was used to determine the type of inhibition between Q7, Q13, Q22, and acarbose and α-glucosidase (Wu et al., 2019; Zeng et al., 2016).

2.4. Molecular docking study

AutoDock Vina (v1.5.7) was used to investigate the binding between α-glucosidase and the inhibitor. The α-glucosidase crystal structure was obtained from the Protein Data Bank (PDB ID: 3A4A). After adding hydrogen, the sample was prepared using AutoDock Vina Tools (Yu et al., 2020). Compounds Q7, Q13, Q22, and acarbose were modeled using Chem 3D. The docking box for 3A4A had center coordinates of 23.244, −7.482, and 23.571, and grid points of 40 × 40 × 40. While α-glucosidase was rigid, Q7, Q13, Q22, and acarbose were flexible. The docking result with the lowest energy was considered optimal. Specific parameters were adjusted to ensure different conformations. The results were analyzed using AutoDock Vina Tools, and molecular interactions were assessed using Discovery Studio.

2.5. Molecular dynamics simulation

Molecular dynamics (MD) simulations were performed using GROMACS 2020.3, applying the amber99sb-ildn force field for proteins and GAFF for ligands. The proteins were placed in a box at a distance of 1.0 nm and solvated. $Cl^{-}$and Na^{+} ions were added for neutralization, and the system was energy-minimized with the steepest descent of 5.0×10^{4} steps. It underwent two equilibrium phases: NVT at 300 K for 100 ps and NPT at 1 bar for 100 ps. The main MD simulation lasted 100 ns at 300 K and 1 atm using the V-rescale and Parrinello–Rahman methods for the temperature and pressure. The Lennard–Jones potentials processed van der Waals forces, LINCS algorithm constrained bond lengths, and the

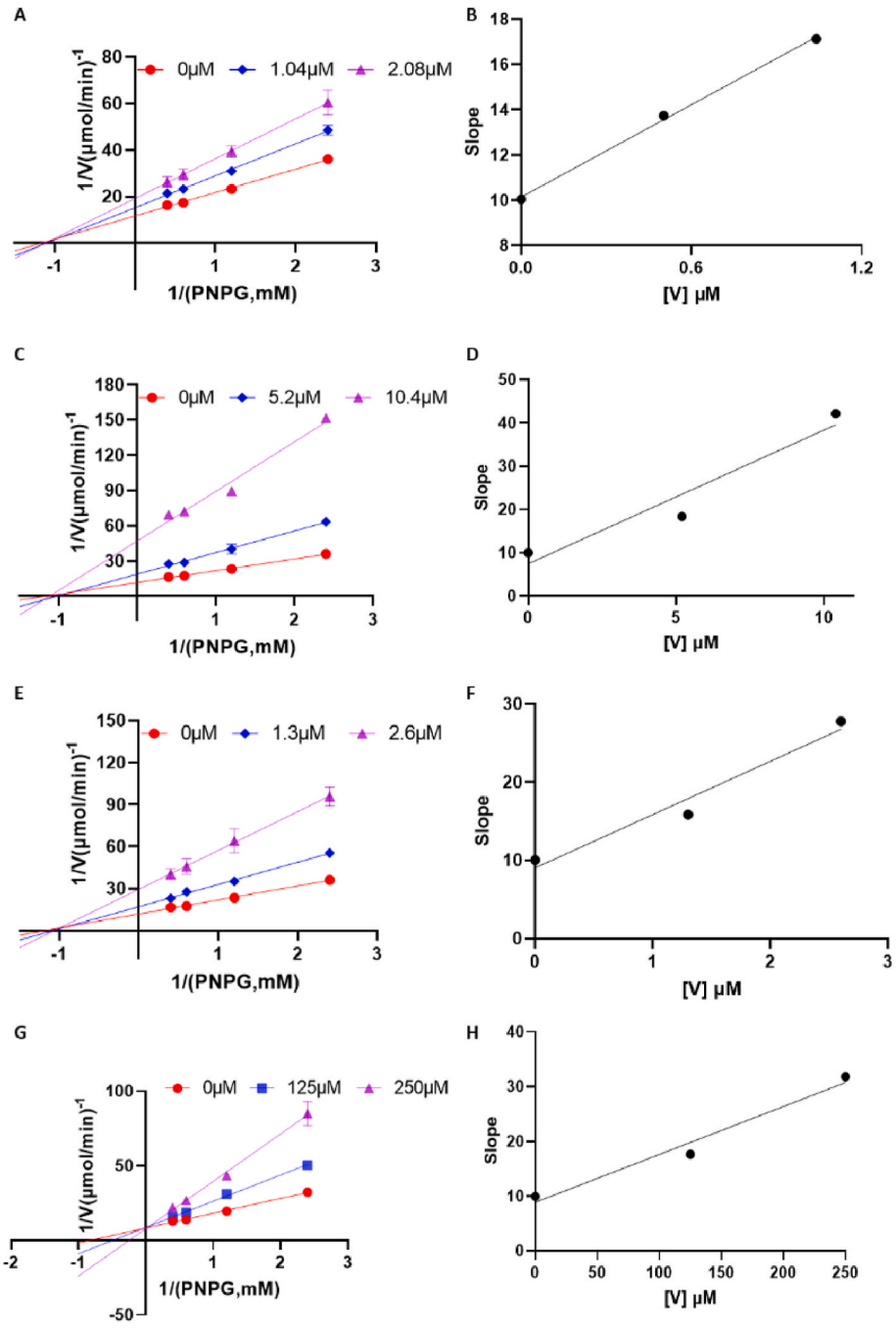


Fig. 1. Kinetic study of Q7, Q13, Q22, and acarbose versus α-glucosidase. (a) Q7 Lineweaver–Burk diagram; (b) Slope versus inhibitor plot of Q7 to evaluate the K_i constant; (c) Q13 Lineweaver–Burk diagram; (d) Slope versus inhibitor plot of Q13 to evaluate the K_i constant; (e) Q22 Lineweaver–Burk diagram; (f) Slope versus inhibitor plot of Q22 to evaluate the K_i constant; (g) Acarbose Lineweaver–Burk diagram; (h) Slope versus inhibitor plot of acarbose to evaluate the K_i constant.

Ewald method particle network was used for long-range electrostatics (Azam et al., 2012; Zheng et al., 2023).

2.6. Model and treatment of diabetic mice

The animal experiments were approved by the Animal Experimentation Ethics Committee of Wuhan University (protocol number: SCXK (E) 20,210,021). Male C57BL/6 mice weighing 18–20 g were purchased from the Wuhan Animal Disease Control Center. Upon arrival, they were allowed ad libitum access to food for one week to acclimate. They were then randomized; one group received a standard diet (control), whereas the other received a high-fat diet for five weeks. After 12 h of fasting, mice in the high-fat diet group were intraperitoneally injected with STZ (80 mg/kg). One week after injection, blood glucose levels were determined from tail tip samples, and mice with values above 11.1 mM were identified as diabetic. Diabetic mice were further divided into different treatment groups: Model, Q7, and metformin (Met), all of which received weight-adjusted doses of 150 mg/kg. All treatments were administered in 0.5% sodium carboxymethyl cellulose for four weeks.

To comprehensively assess the therapeutic effect of Q7 in T2DM mice, we measured the serum glycated protein (GSP) levels. GSP serves as an indicator of the average blood glucose level over the last 1–3 weeks and provides an overview of the current glycemic control (Dong et al., 2020).

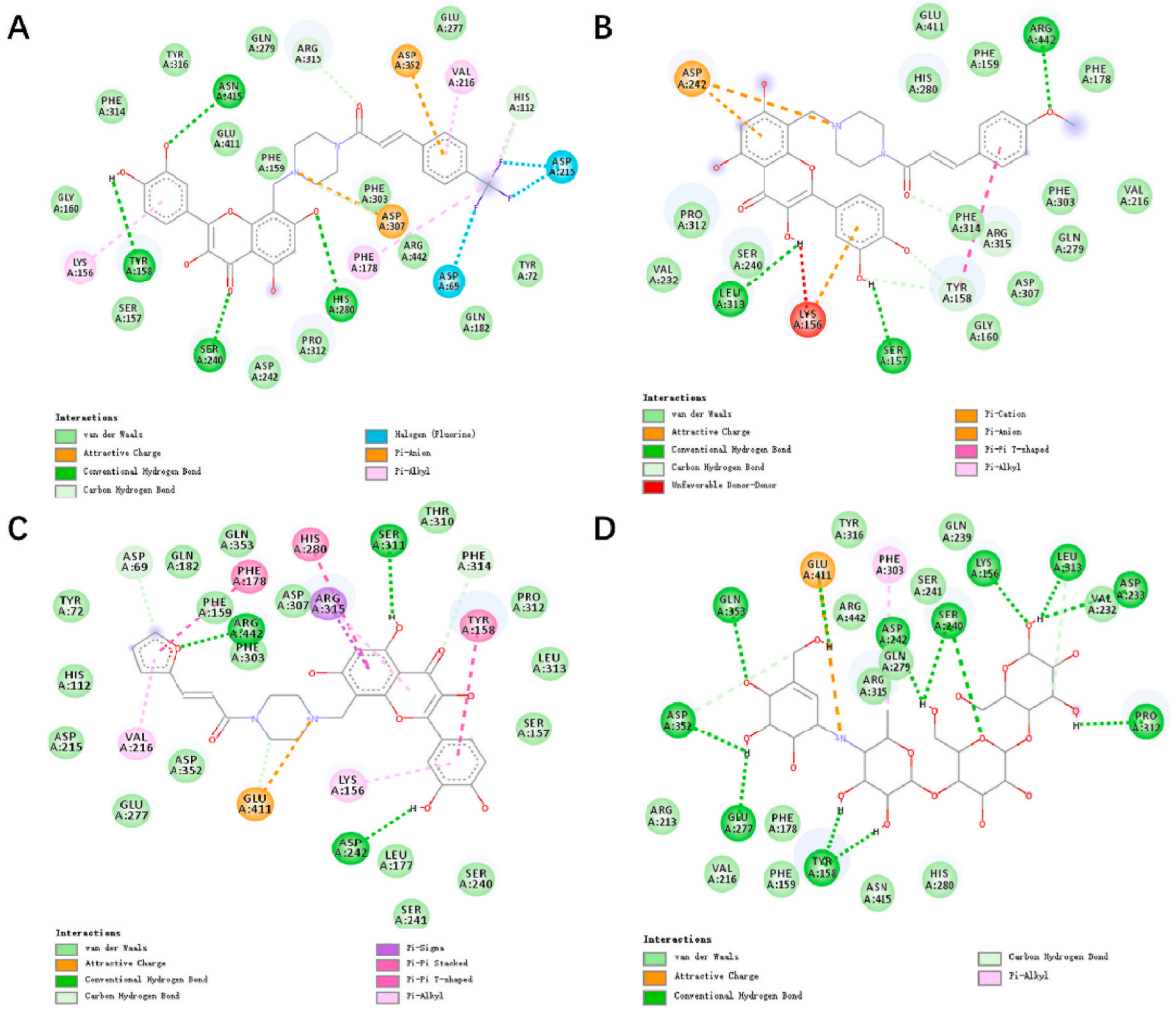


Fig. 2. Docking study of the interaction of the binding pocket of α-glucosidase with (a) Q7; (b) Q13; (c) Q22; and (d) acarbose.

2.7. Microbiota analysis

High-throughput *16S rDNA* sequencing of stool samples was performed by OE Biotech Co. Ltd. (Shanghai, China). Genomic DNA was extracted from the stool samples using a QIAamp 96 PowerFecal QIAcube HT Kit (Qiagen, Hilden, Germany). The V3–V4 regions of the *16S rRNA* genes were amplified by polymerase chain reaction (PCR) using the bacteria-specific primers 343 F (5′-TACGGRAGGCAGCAG-3′) and 798 R (5′-AGGGTATCTAATCCT-3′). Amplicons were purified and sequenced using an Illumina NovaSeq 6000 system.

We systematically observed ten microbial genera, including Muribaculaceae, Lachnospiraceae_*NK4A136_group, Bacteroides, Colidextribacter, Parabacteroides, Blautia, Mucispirillum, Lachnoclostridium, Roseburia, Blautia,* and *[Eubacterium]_coprostanoligenes_group* to investigate the possible effects of drug therapy on gut microbiota structure.

2.8. Statistical analyses

All data are presented as mean ± standard deviation (SD). Differences between groups were determined using repeated one-way analysis of variance (ANOVA). Although $P < 0.05$ was generally used, we have also used $P < 0.01$ or $P < 0.001$ for some of the data to indicate the greater significance of some of the differences. Statistical tests were performed using Graphpad Prism 9.5 software.

3. Results

3.1. Inhibitory activity assays for α-glucosidase

The evaluation of α-glucosidase inhibitory activity revealed that the IC_{50} value for quercetin was 7.97 ± 0.89 μM, which was significantly higher than that of the synthesized quercetin analogs. These results provide strong evidence for the success and rationality of modifying quercetin, as the modified compounds showed superior performance in inhibiting α-glucosidase.

Next, the synthesized Q1–Q23 compounds were tested for their inhibitory activity against α-glucosidase. Acarbose served as a positive control drug (IC_{50} = 124.22 ± 4.48 μM) (Table 1), and the α-glucosidase inhibitory effect of Q7 compounds was 100 times greater than that of acarbose. A comparison of compounds Q2–Q6 revealed that the –H atom was replaced by an electron-withdrawing –F atom of a similar size. Q2 exhibited the highest activity. The introduction of a substituent at the para position of the benzene ring enhanced the α-glucosidase inhibitory activity. For Q7, replacing the –H atom with an electron-withdrawing $–CF_3$ group significantly improved its activity. The stronger the electron-withdrawing ability of the benzene ring, the better the activity. This compound was identified as the most promising compound. Replacing the –H atom with an electron-withdrawing halogen atom (-F, -Cl, -Br) increased the inhibition of α-glucosidase, following the -Cl, -Br, –F trend. Conversely, the inhibitory effect decreased when electron-donating groups, such as $–CH_3$ or $–OCH_3$ found in Q11 and Q13–Q20

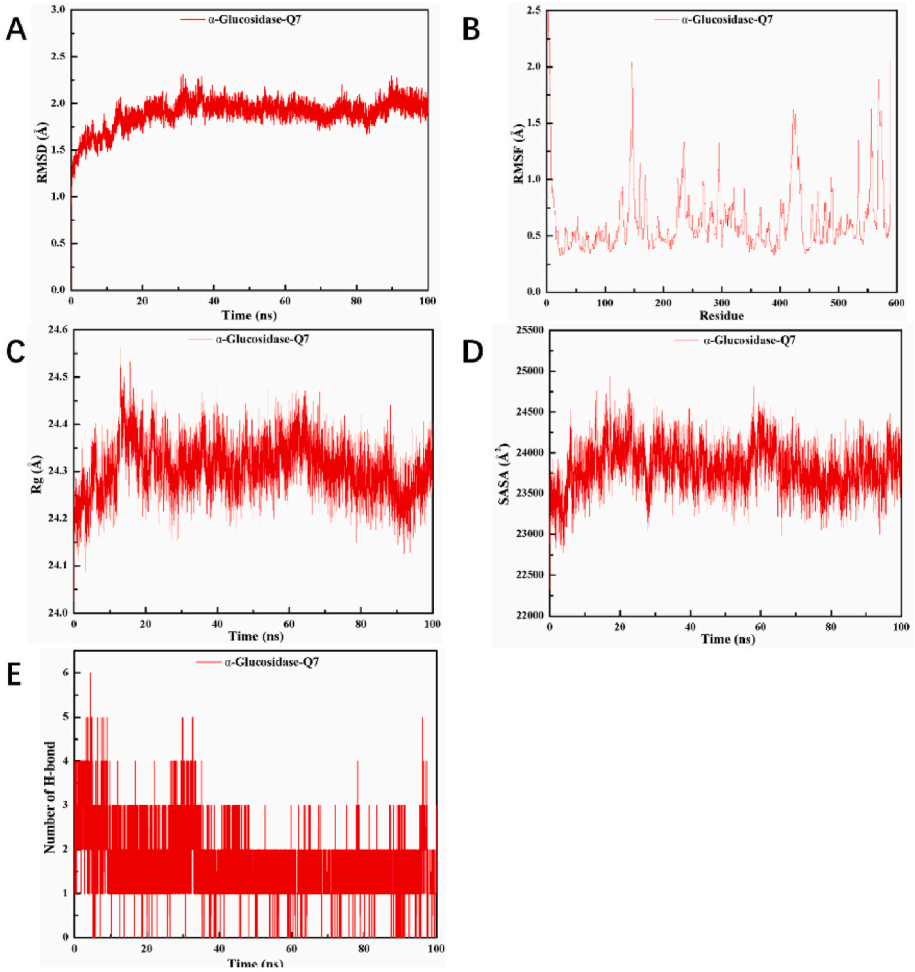


Fig. 3. MD simulation plots for α-glucosidase with ligand Q7 over 100 ns. (a) RMSD plot; (b) RMSF trajectory; (c) Rg over time; (d) SASA diagrams of complexes; (e) Hydrogen bonds over time.

replaced the –H atom. In general, the electron-withdrawing groups on the benzene ring were more effective than the electron-donating groups in inhibiting α-glucosidase activity. Interestingly, compounds Q22 and Q23 showed the opposite results when the benzene ring was replaced with furan or thiophene, with the furan group exhibiting better inhibitory activity than the thiophene group. These results suggest that the molecular size of the analogs may play an important role in their interaction with α-glucosidase. Overall, all target compounds showed superior inhibitory activities against α-glucosidase, suggesting that quercetin analogs are promising as new α-glucosidase inhibitors.

3.2. Kinetics of α-glucosidase inhibition

Enzyme kinetic studies with different substrate concentrations Q7, Q13, Q22, and acarbose showed the inhibition type and constants (K_i) of these compounds versus α-glucosidase using Lineweaver–Burk and Dixon plots (Brindis et al., 2011). The Lineweaver–Burk plot showed a non-competitive mode of inhibition of Q7, Q13, and Q22, whereas acarbose was competitive (Fig. 1). The Dixon plot showed an excellent linear relationship between the slope and Q7, Q13, Q22, and acarbose concentrations (Fig. 1g). The secondary curve, in which the slope was plotted against the Q7, Q13, Q22, and acarbose concentrations, determined the K_i of the enzyme inhibitor. The K_i values for Q7, Q13, Q22, and acarbose were 1.02, 2.42, 1.32, and 101.60 μM, respectively.

3.3. Docking studies

To illustrate the binding modes of quercetin analogs at the α-glucosidase binding pocket, we performed molecular docking studies with the representative compounds Q7, Q13, Q22, and acarbose. The results indicate that the optimal binding energies of α-glucosidase with Q7, Q13, Q22, and acarbose are −11.6, −10, −10.9, and −8.9 kcal/mol, respectively. Therefore, Q7 had the strongest binding affinity for α-glucosidase, which was consistent with the experimental results.

The interactions of Q7, Q13, and Q22 with α-glucosidase revealed different interaction types that elucidated their binding modes in the binding pocket of the enzyme. Q7 demonstrated complexity and diversity at the molecular level through specific interactions with multiple amino acids. van der Waals interactions involve amino acids such as GLY160, PHE314, TYR316, GLU411, PHE159, GLN279, GLU277, TYR72, GLN182, PHE303, ARG442, PRO312, ASP242, and SER157, which contribute to the stability of the molecular structure by non-covalent forces. In addition, hydrogen bond interactions with TYR185, ASN415, HIS280, and SER240, and halogen interactions with ASP215 and ASP69 further strengthened the intramolecular attraction. The π-anion interaction with ASP352 and the attractive charge interaction with ASP307 can lead to charge distribution adjustments and influence the electronic structure and properties of the molecule. The diversity of interactions, including π-alkyl interactions with LYS156, PHE178, and VAL216 and carbon–hydrogen bond interactions with HIS112 and ARG315, highlights the versatility of Q7 in its interactions with proteins and lays the foundation for its potential functions in biological systems

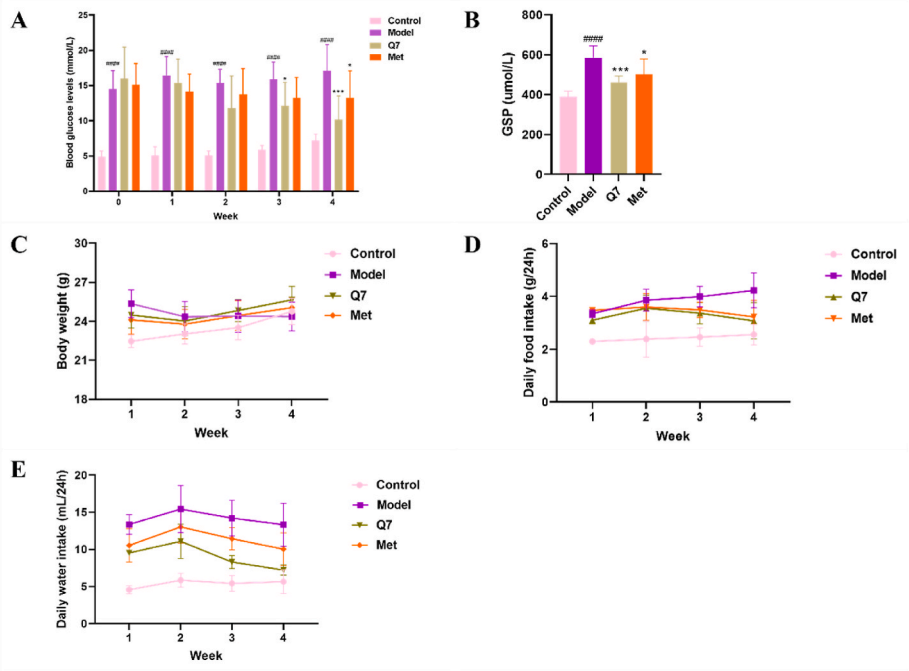


Fig. 4. Effects of Q7 on metabolic parameters in T2DM mice: (a) FBG; (b) GSP; (c) body weight; (d) daily food intake; (e) daily water intake. Data are presented as mean SD (n = 8). $^{\#\#\#\#}P < 0.0001$ vs. Control; $^{*}P < 0.05$, and $^{***}P < 0.001$ vs. Model.

(Fig. 2a).

In Q13, van der Waals interactions were detected between PRO312, SER240, VAL232, GLU411, HIS280, PHE159, PHE178, VAL216, PHE303, GLN279, PHE314, ASP307, and GLY160. Conventional hydrogen bonds were formed with LEU313 and ARG442, whereas attractive charge and π-cation interactions occurred with ASP242. π-Anion interactions occurred in LYS156 and π-π interactions were observed in TYR158. In addition, unfavorable donor–donor interactions were formed with LYS156, and carbon–hydrogen bonds with ARG315 and TYR158 (Fig. 2b).

Q22 showed van der Waals interactions with ASP352, GLU277, ASP215, HIS112, TYR72, GIN182, PHE159, PHE303, ASP307, THR310, PRO312, LEU313, SER157, LEU177, SER240, and SER241. Conventional hydrogen bonds were identified with ARG442, SER311, and ASP242, and π-alkyl interactions were observed with VAL216 and LYS156. In addition, π-π interactions occurred with PHE178, HIS280, and TYR158, and π-sigma interactions with ARG315. Attractive charge interactions were observed with GIU411, and carbon–hydrogen bonds were formed with ASP69 and PHE314. These comprehensive observations deepen understanding of the binding modes and interactions of these compounds with α-glucosidase (Fig. 2c).

Furthermore, multiple interactions involving different amino acids were observed with acarbose. First, acarbose formed van der Waals interactions with ARG213, VAL216, PHE159, PHE178, ASN415, HIS280, VAL232, ARG315, SER241, GLN279, ARG442, TYR316, and GLN239. It also formed conventional hydrogen bonds with GLN353, APS352, TYR158, PRO312, ASP233, LEU313, LYS156, SER240, and ASP242. In addition, it formed an attractive charge interaction with GIU411, π-alkyl interactions with PHE303, and a carbon–hydrogen bond with LEU313 (Fig. 2d).

3.4. Molecular dynamics simulation

During the 100 ns MD simulation, the root means square deviation (RMSD) of α-glucosidase-Q7 initially increased but subsequently stabilized between 1.5 Å and 3 Å, highlighting its structural stability (Fig. 3a). Further analysis revealed an RMSD of 1.89 ± 0.15 Å, providing solid evidence for the structural stability of the protein-ligand complex. The small RMSD fluctuations observed may be related to the transient repositioning of the ligand or short-lived structural adjustments. These results strongly support the stable binding between α-glucosidase-Q7 and its ligand.

According to the root mean square fluctuation (RMSF), the Q7 value with α-glucosidase peaked at 2.04 Å. This suggests that the binding of Q7 to α-glucosidase is robust and causes minimal external interference. Remarkably, Q7 showed good structural stability at conformational sites despite RMSF fluctuations during the 100 ns simulation, supporting its stable binding as a non-competitive inhibitor of α-glucosidase (Fig. 3b).

Furthermore, the radius of gyration (Rg) value for Q7 was 24.30 ± 0.05 Å (Fig. 3c) and the average solvent accessible surface area (SASA) was 23,823 ± 278.20 $Å^2$ (Fig. 3d). Both parameters indicated that the conformation and external solvent exposure of the complex remained stable throughout the simulation.

Another important finding was that the average number of hydrogen bonds between Q7 and α-glucosidase was 1.52 ± 0.79, highlighting the strong and consistent interaction between them. Throughout the simulation, the number fluctuated between 40 and 55, further highlighting the stability of the complexes (Fig. 3e).

In summary, these analytical results clearly demonstrate the structural stability of the α-glucosidase-Q7 complex and its non-competitive inhibitory mechanism and provide compelling evidence for further research on the potential of this compound as a drug candidate.

3.5. Effects of Q7 on metabolism

We employed a high-fat diet/streptozotocin (HFD/STZ) induction method to mimic the metabolic abnormalities characteristic of human T2DM. HFD/STZ-treated mice exhibited significantly increased fasting blood glucose (FBG) levels compared with the control group, which persisted in subsequent observations, confirming the successful establishment of a T2DM mouse model. However, by the fourth week, the mice in the Q7 treatment group showed a significant decrease in FBG levels ($P < 0.001$). In contrast, Met, a commonly used drug for treating T2DM, demonstrated a hypoglycemic effect in the fourth week ($P <$

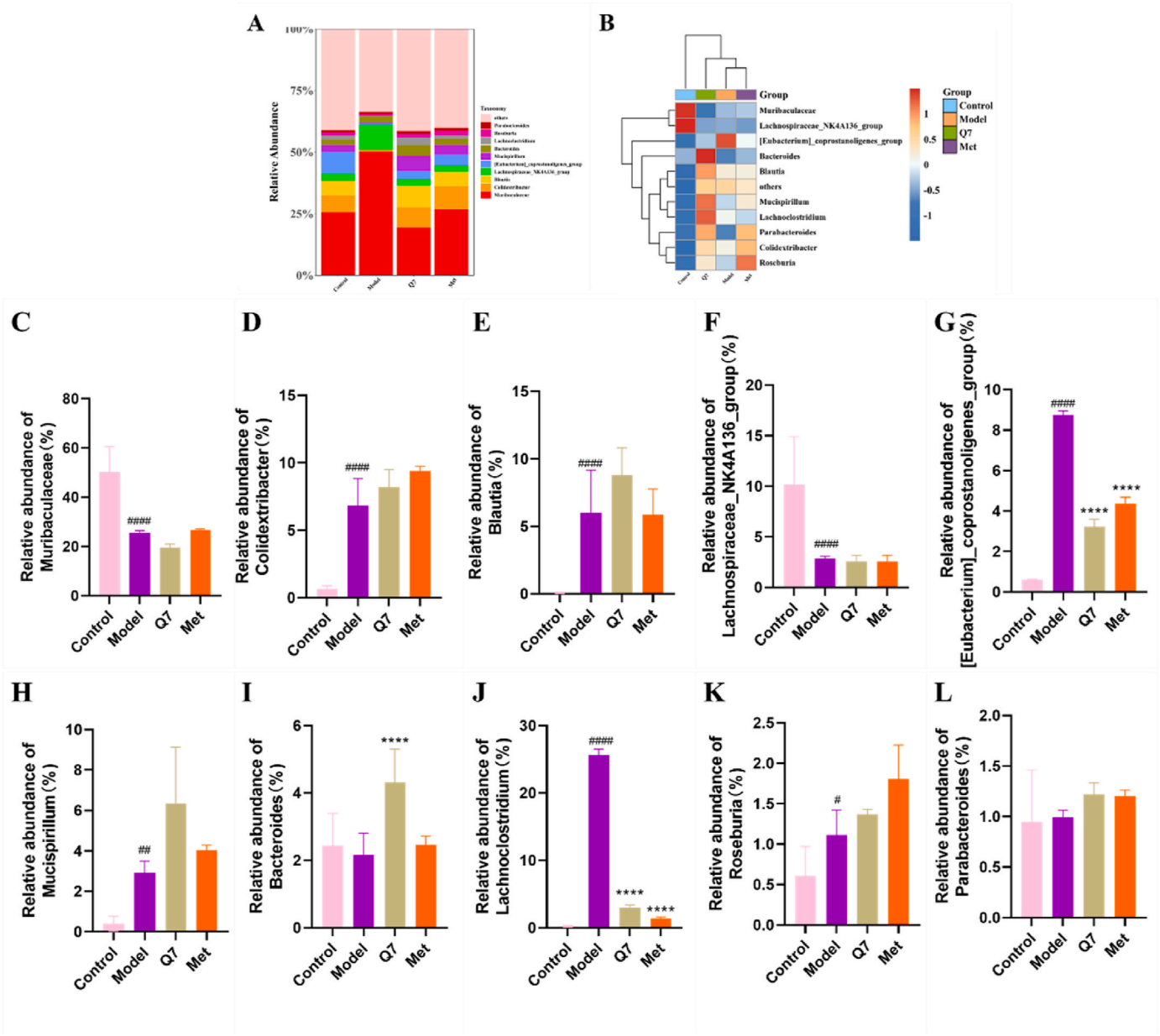


Fig. 5. Effects of Q7 on gut microbiota in T2DM mice. (a) Genus level abundance (Top 10); (b) Genus level heatmap (Top 10); (c) *Muribaculaceae*; (d) *Colidextribacter*; (e) *Blautia;* (f) *Lachnospiraceae_NK4A136_group;* (g) *[Eubacterium]_coprostanol igenes_group;* (h) *Mucispirillum;* (i) *Bacteroides;* (j) *Lachnoclostridium;* (k) *Roseburia;* (l) *Parabacteroides*. Data are presented as mean ± SD (n = 8). $^{\#\#\#\#}P < 0.0001$ vs. Control; $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{****}P < 0.0001$ vs. Model.

0.05) (Fig. 4a). At the end of the experiment, the GSP values in the Q7 treatment group were significantly lower than those in the model group ($P < 0.001$) (Fig. 4b).

During the experiment, mice in the control group gradually increased in body weight. In comparison, the mice in the model group showed weight loss. After intervention with Q7, there was an increase in the body weight of the mice. Furthermore, both Q7 and Met significantly reduced food and water intake in the mice (Fig. 4d and **e**), with the Q7 group showing a more pronounced effect.

In the HFD/STZ-induced T2DM mouse model, Q7 and Met treatments effectively reduced blood glucose and GSP levels, and controlled weight loss and water and food consumption. The effects of H7 were more pronounced than those of Met.

3.6. Effects of Q7 on the intestinal microbiota in T2DM mice

We systematically observed ten microbial genera in the experimental mice. The results showed a significant decreased ($P < 0.0001$) in the relative abundance of *Lachnoclostridium,* and *[Eubacterium]_coprostanoligenes_group* in the Q7 group compared with that in the model group (Fig. 5g and **j**). In addition, the relative abundance of *Bacteroides* significantly increased compared with that in the model group ($P < 0.0001$) (Fig. 5i). However, the Q7 group showed no significant differences in other microbial genera in compared with the model group ($P > 0.05$) and showed either higher or lower abundance than the control group. This suggests that Q7 may not have a pronounced effect on the restoration of the gut microbiota.

4. Results and discussion

Q7, a quercetin analog, exhibits exceptional α-glucosidase inhibitory activity, outperforming both the standard drug acarbose and quercetin. This superior performance was primarily attributed to the presence of an electron-withdrawing CF_3 group on the benzene ring of Q7, which

significantly enhanced its inhibitory effect.

In kinetic analyses, Q7 showed non-competitive inhibition, together with Q13 and Q22. This study showed that Q7, a quercetin analog, has exceptional α-glucosidase inhibitory activity, a mode of action distinct from the competitive inhibition of acarbose. Molecular docking studies confirmed these results and showed that Q7 has a strong binding affinity for α-glucosidase. MD simulations confirmed the structural stability of the α-glucosidase-Q7 complex and highlighted its potential as a potent, non-competitive inhibitor.

In our *in vivo* experiments, we chose Q7 as the treatment group not only because it had the strongest inhibitory activity but also because of its $–CF_3$ group, which increases metabolic stability and extends the half-life of the compound. Furthermore, the $–CF_3$ group can alter molecular lipophilicity, thereby improving the penetration of biological membranes (Roy et al., 2011). As a positive control, we chose Met, which, according to literature, is the preferred drug for patients with T2DM (Arun et al., 2020; Dong et al., 2011; Liu et al., 2023).

The effectiveness of Q7 was demonstrated by the significant reduction in FBG levels in diabetic mice, which was more significant than that of Met. The levels of GSP, another indicator of blood sugar levels, also decreased in both the Q7 and Met groups. However, this reduction was more significant in the Q7 group ($P < 0.001$), indicating its superior efficacy in glucose regulation compared with Met ($P < 0.05$).

The effect of Q7 on intestinal microbiota was also notable. A dose of 150 mg/kg of Q7 resulted in carbohydrate accumulation in the colon, which may alter the composition and function of the gut microbiota. Certain microbes known to regulate the gut microbiota and mitigate inflammatory processes in T2DM, such as Muribaculaceae, *Mucispirillum*, and *Parabacteroides* (Xie et al., 2023; Zhu et al., 2023), and those that produce short-chain fatty acids, such as *Bacteroides* and *Blautia*, were affected (Li et al. 2020, 2023; Wan et al., 2019). In particular, in the Q7 group, genera such as *Bacteroides*, *[Eubacterium]_coprostanoligenes_group*, and *Lachnoclostridium* showed significant changes compared with the model group ($P < 0.0001$).

5. Conclusions

Q7 is a promising α-glucosidase inhibitor in T2DM treatment, outperforming conventional treatments such as Met. However, the possible accumulation of carbohydrates in the colon at a dose of 150 mg/kg requires further investigation to ensure maximal therapeutic benefit, while minimizing the impact on gut microbiota function. This comprehensive analysis highlights the need for precise dosage management for the clinical use of Q7, considering its therapeutic potential and effects on intestinal health.

CRediT authorship contribution statement

Honghui Liu: Writing – original draft. **Yan Wang:** Data curation. **Jing Tong:** Writing – review & editing. **Jing Li:** Writing – review & editing. **Hong Ding:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fbio.2024.103713.

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